

Astroinformatics 1: Graded Practice 3

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I. LIGHT CURVES PLOTTING

The main objective of this practice was to improve the light curve plots from Practice 1. In order to do so, we worked directly with the CSV files generated in Practice 1, as it was easier to read and import the data into Data-Frames. Lets remember that there were 20 files in total, all of which originally came from FITS files. Also, the files were modified to keep the most important information needed for the light curve plot. Therefore, only the following columns were kept:

- TIME
- TIMECORR
- SAP FLUX
- SAP FLUX ERR
- PDCSAP FLUX
- PDCSAP FLUX ERR

Considering that we had the errors on the measurements, it was decided that the best way to plot the light curves with as much information as possible was to use the function `matplotlib.pyplot.errorbar`.

II. OUTLIER DETECTION

The following task of the activity was to showcase in the light curve plots the outlier data. In order to do so, it was decided that an outlier would be defined as any datapoint with a separation of 3σ away from the median of the data. The following section of code was used to identify the data outliers:

```
t = np.asarray(df[time_col])
y = np.asarray(df[flux_col])
dy = np.asarray(df[err_col])

mask = np.isfinite(t) & np.isfinite(y)
      & np.isfinite(dy)
t = t[mask]
y = y[mask]
```

```
dy = dy[mask]
```

```
# Robust outlier detection
sigma_outlier = 3
med = np.median(y)
mad = median_abs_deviation(y)
robust_sigma = 1.4826 * mad
outliers=np.abs(y - med)>
           sigma_outlier*robust_sigma
good = ~outliers
```

Additionally, to differentiate the outliers from the rest of the data in the light curve, besides using another color for the marker, it was decided to use stars on the markers and to add horizontal caps in their error bars, also the transparency of this error bars was set up to be higher than of the rest of the data so it could stand up more.

III. STATISTICS

Following the previous idea to define the outliers, it was decided to use both the median and 3σ limits as parameters to characterize each light curve, and add them to the plot as horizontal lines (with different colors and line-styles). Also, the other parameter used to characterize each light curve was their periods. In order to obtain them, a Lombscargle method was implemented in the following section of code:

```
# Lomb-Scargle
baseline = t.max() - t.min()

if min_period is None:
    min_period =
        2 * np.median(np.diff(np.sort(t)))
if max_period is None:
    max_period = baseline / 2

min_freq = 1 / max_period
max_freq = 1 / min_period

ls = LombScargle(t[good], y[good], dy[good])
frequency, power = ls.autopower(
    minimum_frequency=min_freq,
    maximum_frequency=max_freq,
    samples_per_peak=oversampling
)
```

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```
best_frequency = frequency[np.argmax(power)]
best_period = 1 / best_frequency
```

```
# Phase
```

```
phase = (t / best_period) % 1
```

As we can observe on the last line of code, the period allowed us to transform the data from being expressed in Time-space into Phase-space, which improves the readability of the light curve. Now, we present some of the most interesting light curves obtained, along with their periodograms from the Lombscargle algorithm:

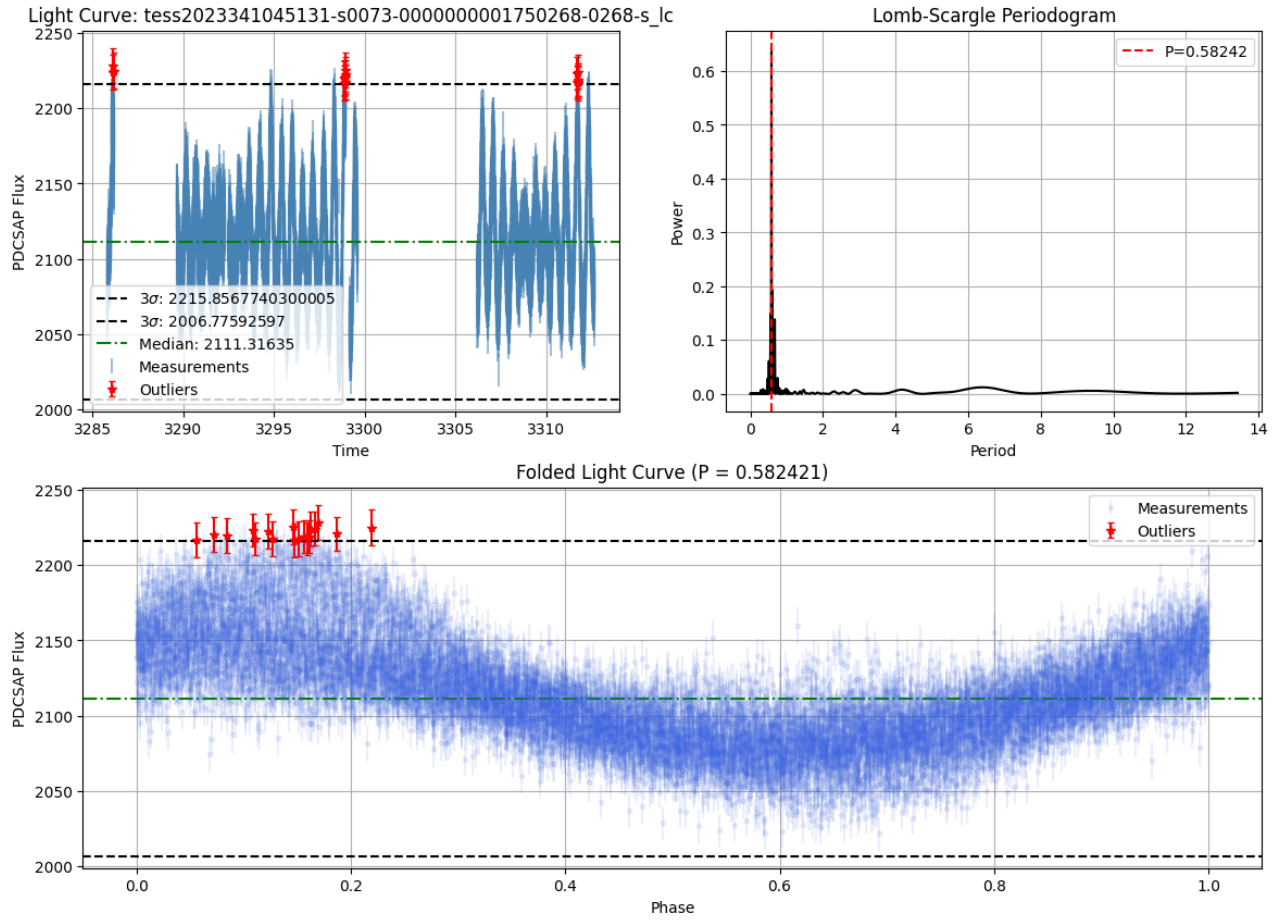


Figure 1. Light curve with sinusoidal form.

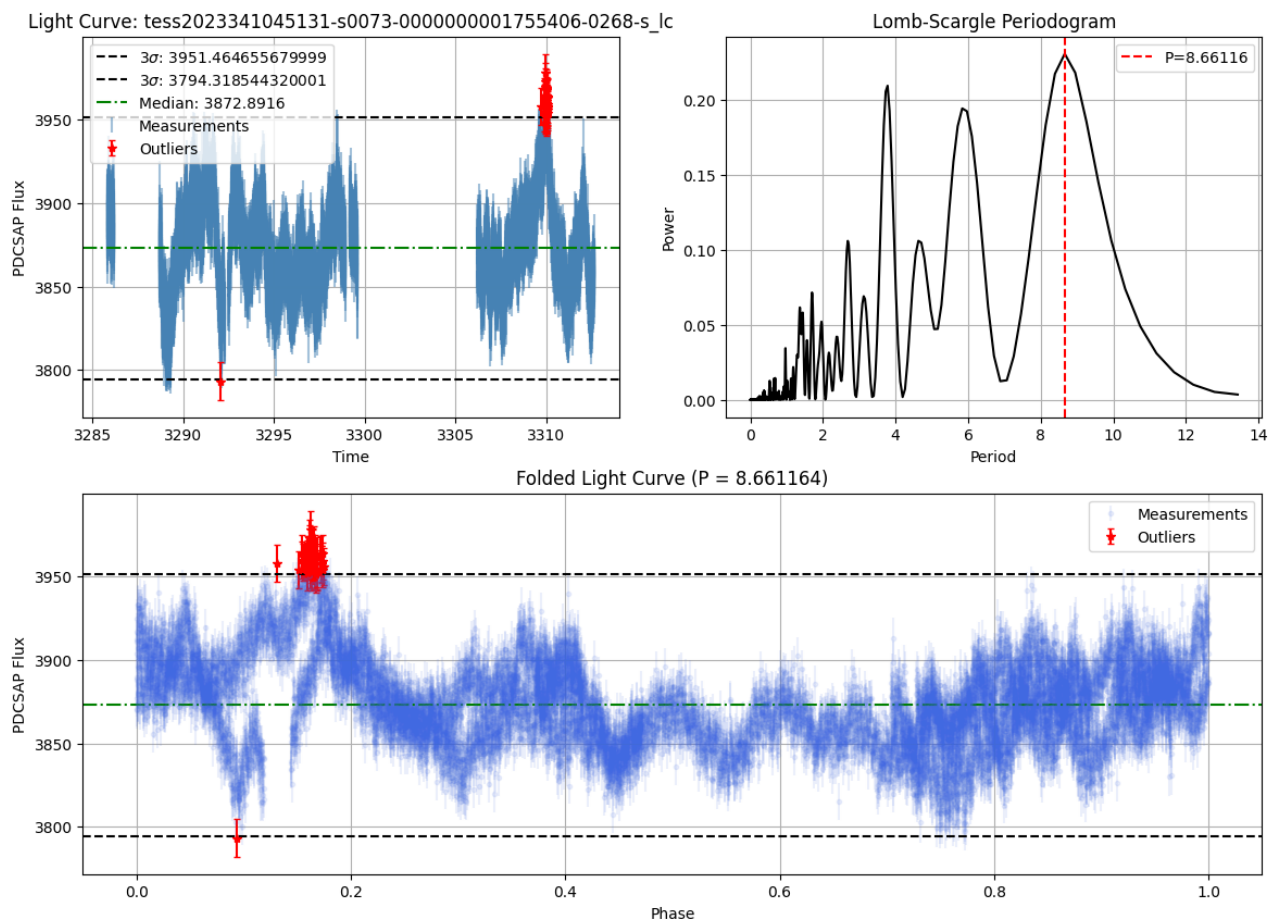


Figure 2. Light curve with irregular form.

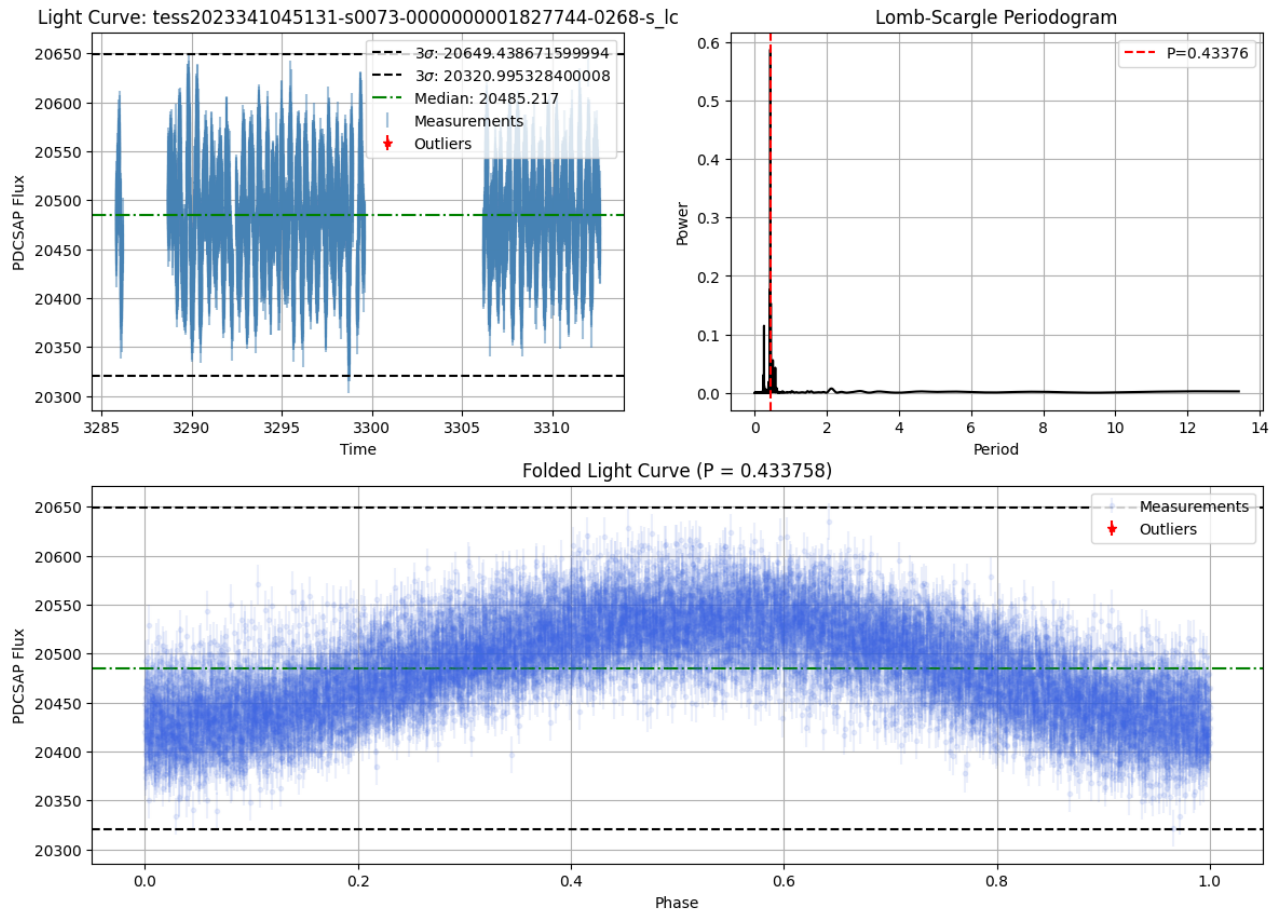


Figure 3. Light curve with cosine form.

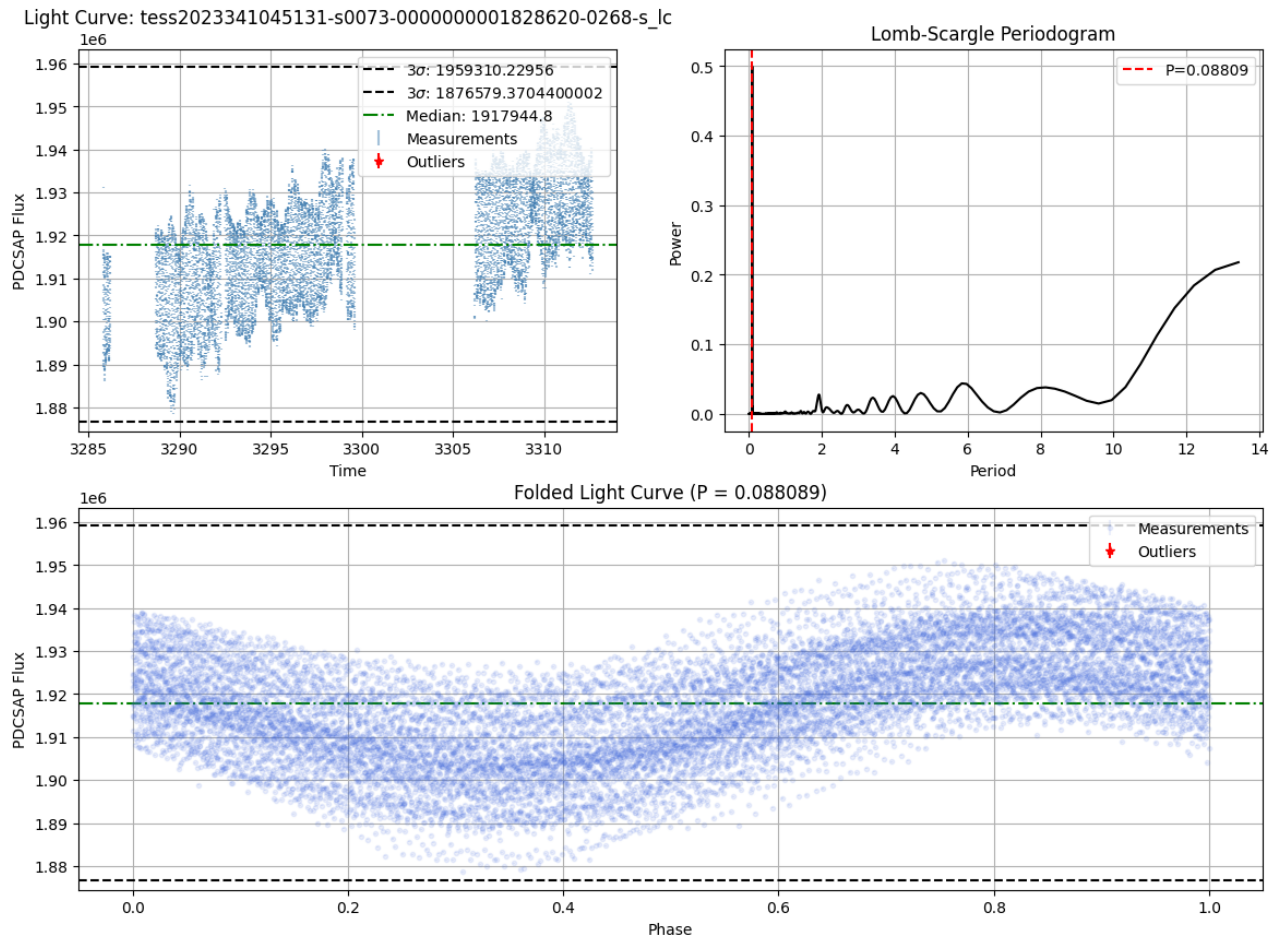


Figure 4. Light curve with weird form.

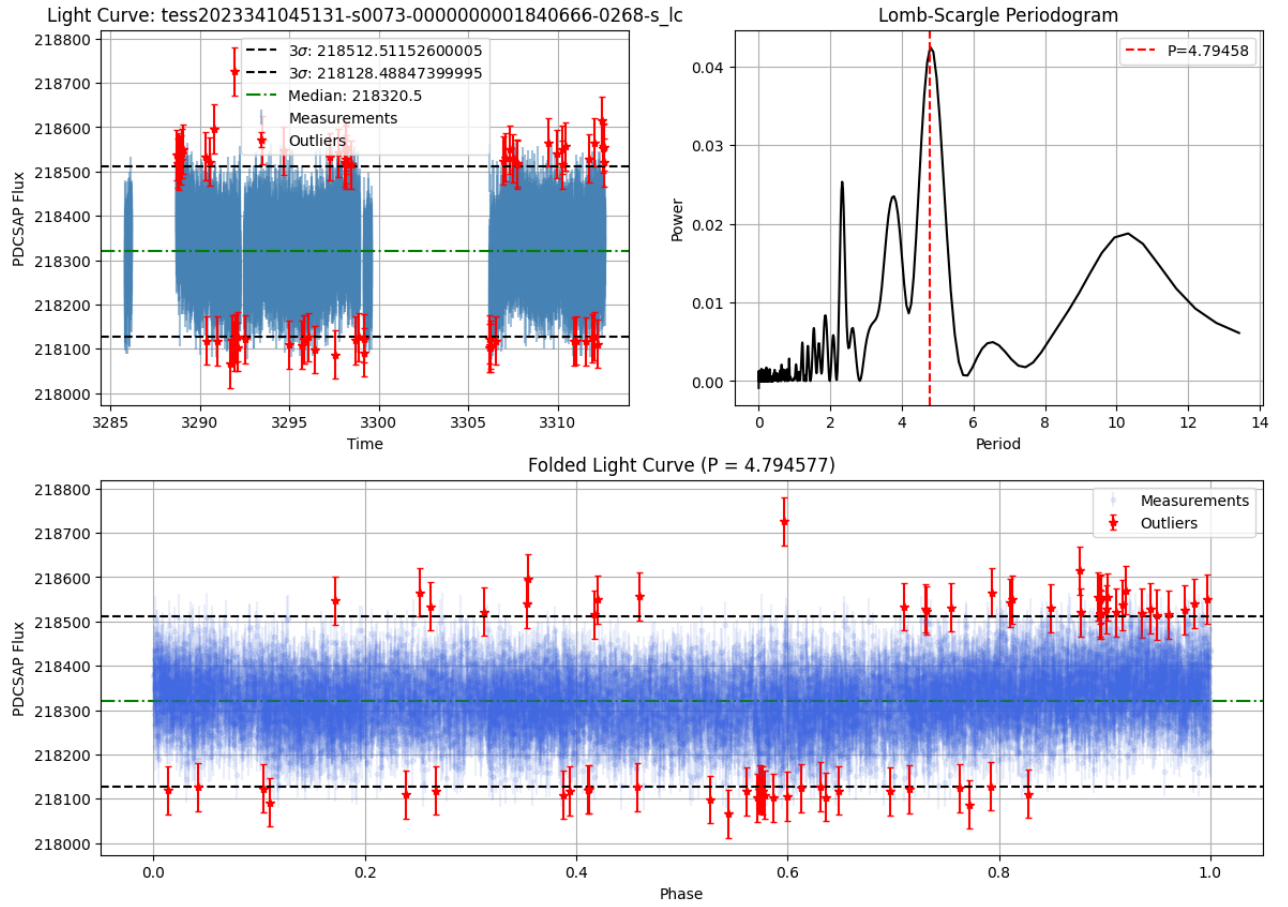


Figure 5. Light curve with no clear variability. It should be noted that the rest of the obtained light curves were extremely similar to this one, so they are not presented here.